Week 7

COGS 108 SP 2026

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Due dates

- Data Checkpoint: Wednesday (05/13)
- D6: Friday (05/15)

Data Checkpoint

- Address proposal feedback in the checkpoint
- Proposal grade will be adjusted if issues are addressed

Models: Evaluation Metric

Metric	Penalizes Complexity?	Main Purpose	Key Idea	Better Value
Adjusted R-squared	Yes	Compare regression models with different numbers of predictors	Adjusts R^2 by penalizing unnecessary variables	Higher is better
R-squared	No	Measure how much variance the model explains	Usually increases when predictors are added, even if they are useless	Higher is better
BIC	Strongly	Model selection and overfitting control	Strong penalty for adding more parameters	Lower is better
Durbin-Watson	No	Check residual independence	Detects autocorrelation in residuals, not model complexity	Closest to 2 is best

Models: Regression Interpretation

Focus on what each part of a fitted model means.

Model form

outcome = intercept + slope × predictor

- The intercept anchors the model.
- The slope represents the expected change in the outcome for a one-unit change in the predictor.
- Multiple regression interprets one predictor while holding the others constant.

Interpretation habit

- Name the outcome first.
- Name the predictor second.
- State the unit of change.
- Avoid causal language unless the design supports it.

Models: Transformations & Diagnostics

Think about why a model may or may not be interpretable.

Transform	Diagnose	Sanity check

- Transformations can make skewed variables easier to model.
- Diagnostics ask whether model assumptions are plausible.
- Sanity checks ask whether the variables and data-generating story make sense.

A low-quality measurement can make a statistically neat model useless.

Model Complexity: Fit vs. Simplicity

More predictors can improve apparent fit, but not always the model you should prefer.

In-sample fit

- Some metrics mostly ask: “How close did the model fit the observed data?”
- These metrics can improve when the model becomes more flexible.

Complexity penalty

- Other metrics ask: “Did the better fit justify the extra parameters?”
- These metrics help discourage unnecessary complexity.

Fit

better prediction of seen data

vs

Simplicity

fewer unnecessary parameters

Dimensionality Reduction: PCA Core Ideas

PCA turns many numeric variables into fewer components.

Principal Component Analysis (PCA)

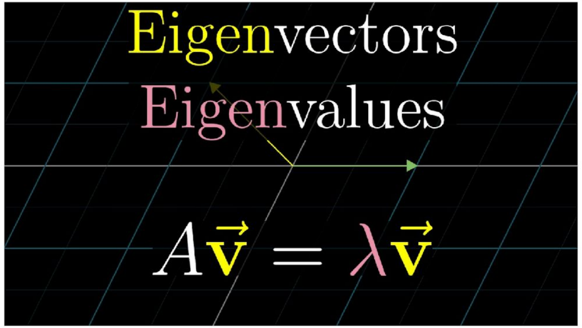
Key Terms:

- **Principal Component (PC)** - a linear combination of the predictor variables
- **Loadings** - the weights that transform the predictors into components (aka weights)
- **Screeplot** - Variance explained of each component

- A principal component is built from the original variables.
- Loadings are the weights used to form each component.
- Components are ordered by how much variance they capture.
- PCA is linear, so it works best when linear projections are useful.

PCA Workflow: From Data to Components

The workflow matters because PCA is sensitive to scaling and centering.



PCA

1. Center the data
2. Compute covariance matrix
3. Compute eigenvectors and eigenvalues of covariance matrix
4. (Optional) subset eigenvectors by picking the m largest eigenvalues
5. Project datapoints into PCs by multiplying with eigenvectors

PCA doesn't work if the data isn't zero mean!

Typically we use a z-score if data is close to normally distributed

PCA Workflow: From Data to Components

The workflow matters because PCA is sensitive to scaling and centering.

How many PCs to select?

Option 1: Visually through the screeplot: elbow method

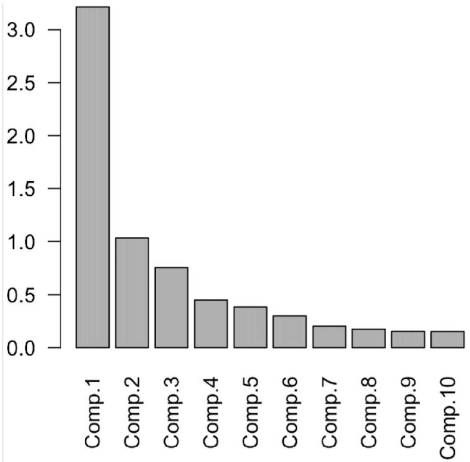


Figure 7-2. A screeplot for a PCA of top stocks from the S&P 500

Option 2: % Variance explained (i.e. 80% variance explained)

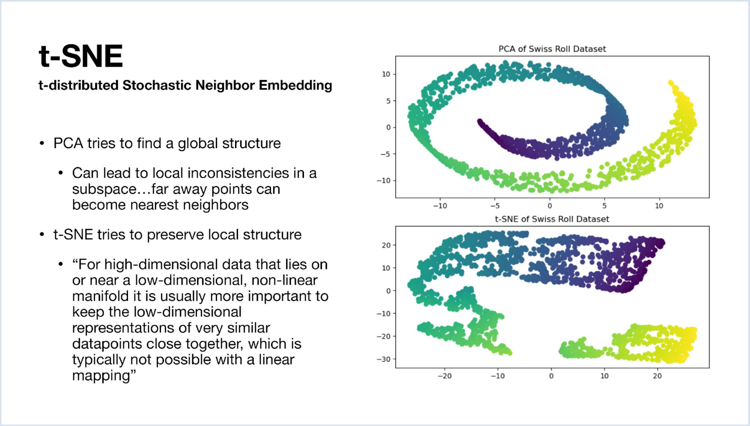
Option 3: Inspect loadings for an intuitive interpretation

Option 4: Cross-validation

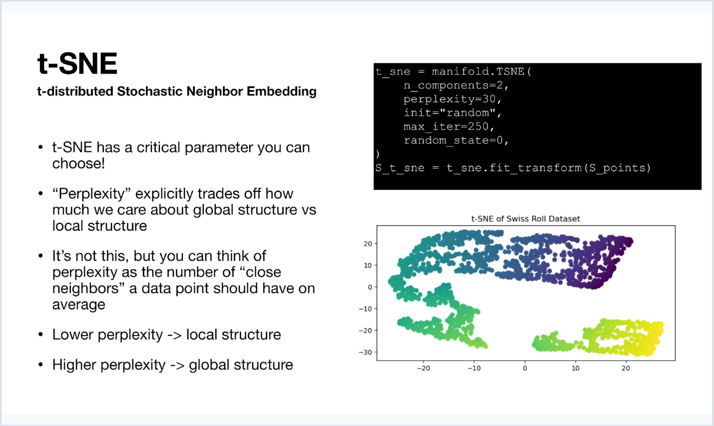
Source: Practical Statistics for Data Scientists (Bruce & Bruce)

Manifold Learning & t-SNE

When linear projection is not enough, local neighborhoods become the focus.



t-SNE local-structure idea



Perplexity tradeoff

- Manifold learning looks for lower-dimensional structure inside high-dimensional data.
- t-SNE emphasizes nearby points staying nearby.
- Perplexity changes the local-versus-global emphasis.
- Visual clusters should be interpreted carefully.

Machine Learning Workflow

Prediction requires a workflow that protects against overfitting.

1. Data Partitioning

- Split data into:
    - Training set, Validation set, Test set
- Prevents information leakage

2. Feature & Model Selection

- Select useful features
- Choose appropriate model architecture
- Decisions should rely only on training/validation data

3. Hyperparameter Tuning

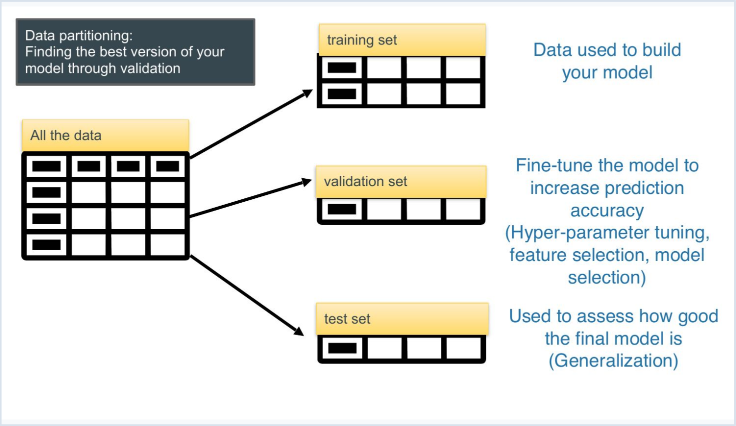
- Optimize parameters such as:
    - learning rate
    - number of neighbors (K)
    - tree depth
- Never tune directly on the test set

4. Model Assessment

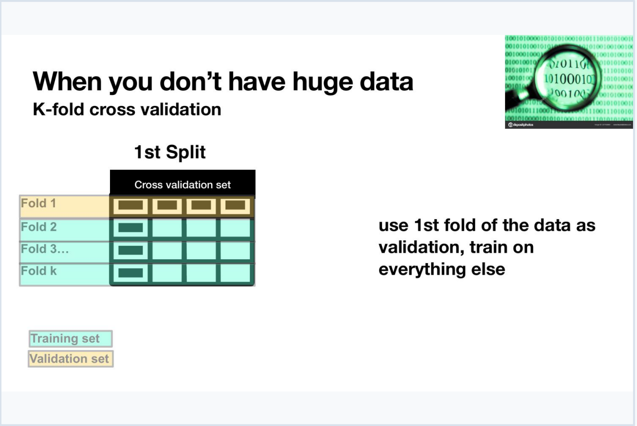
- Evaluate final performance on held-out test data
- Measures generalization to unseen samples

Data Partitioning & Cross-Validation

Separate model building, model tuning, and final model assessment.



Train / validation / test split



K-fold cross-validation

- Training set: fit the model.
- Validation set: tune or choose among models.
- Test set: estimate final generalization.
- K-fold CV reuses limited data more efficiently by rotating the validation fold.

Learning Modes & Classification Metrics

Before choosing a metric, identify the task and the error you care about.

Supervised Learning

- Use labeled data
- Learns relationship between features and known targets
- Goal is prediction/classification
- Has ground truth labels during training
- Examples: Linear Regression, Logistic Regression, SVM, Random Forest

Unsupervised Learning

- Use unlabeled data
- Finds hidden structure or patterns in data
- Goal is discovery, grouping, or dimensionality reduction
- No ground truth labels provided
- Examples: K-means, PCA, t-SNE, Hierarchical Clustering

Supervised learning learns from known answers.
Unsupervised learning tries to discover patterns without answers.

Learning Modes & Classification Metrics

Before choosing a metric, identify the task and the error you care about.

Classification metric questions

- Are the labels known during training?
- Is the task classification or regression?
- Are the classes balanced?
- Which mistake is more costly: false positive or false negative?
- Would raw accuracy hide poor performance on the rare class?

Metric choice should match the decision problem.

Academic Integrity Reminder for Quizzes

- Quiz activity and LockDown Browser logs are **monitored.**
- **Suspicious activity has been recorded.**
- **Do not cheat in any form**, including:
  - using AI/notes
  - sharing codes or answers
  - communicating with others
  - taking the quiz outside section.
- **Further violations will be reported to the Academic Integrity Office.**

Quiz Time

NO Phone, NO Communication, No Notes

Access Code: springsteen

Your time to …

- Work on projects
- Finish Checkpoint
- Finish D6
- Have fun…